# FSPC-Based Spike Compression for Privacy-Preserving Event-Based Edge AI

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## 1. Summary of the Research

- Event-based cameras capture changes in brightness rather than conventional images, offering potential privacy advantages for Edge AI.
  Spiking Neural Networks (SNNs) are energy-efficient neural networks, but spike communication between layers can cause significant overhead.
  Frequent Spike Pattern Compression (FSPC) reduces communication overhead by compressing frequently occurring spike patterns.

- This study aims to develop privacy-preserving and efficient Event-Based Edge AI using SNNs and evaluate FSPC by comparing classification accuracy and spike communication before and after compression.

## 2. Approach/Methodology

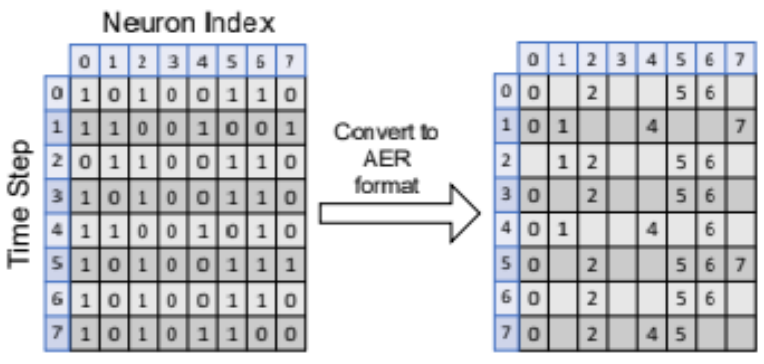
**Experimental Workflow**

1. Train an SNN using an existing event-based dataset (e.g., THU-EACT-50).
2. Evaluate the trained SNN using event data captured by the GENX320 camera.
3. Apply FSPC to the SNN
4. Compare classification accuracy and spike communication before and after compression.

**Frequent Spike Pattern Compression(FSPC)[1]**

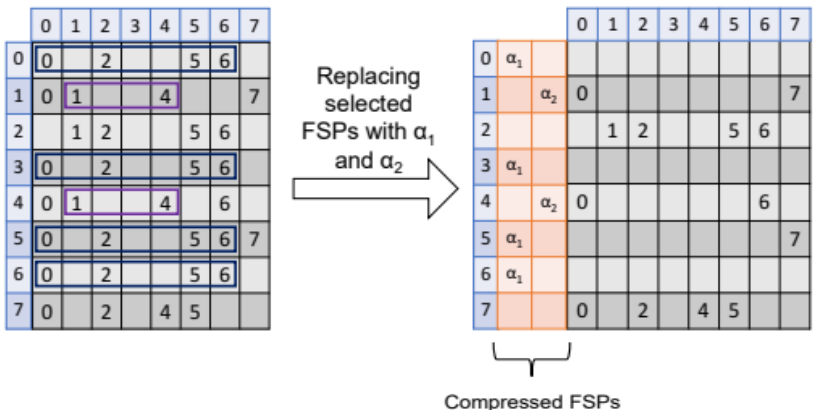
Phase 1: Encode Spike Data to Address Event Representation (AER) Data

Spike Data AER Data




Phase 2: Extract frequent spike patterns(FSPs). assign symbols, and compress them in the AER data.

FSPs in AER Compress FSPs to Symbols




Phase 3: Decompress symbols to recover FSPs, decode AER back to a spike tensor, and forward to the next layer.

## 3. Current Results and Status

**SNN Classification Experiment**

Used the THU-EACT-50 event-based dataset for binary human action classification

| | |
|---|---|
| Test Accuracy | 95.96% |
| Precision | 0.2500 |
| Recall | 0.4167 |
| F1-score | 0.3125 |

## 4. Remaining Tasks and Tentative Schedule

| Month | 10 - 11 | 12 | 1 - 2 |
|---|---|---|---|
| Task | 1 | 2 | 3 |

1: Train a baseline SNN using THU-EACT-50. Test GENX320 event data
2: Evaluate FSPC.
3: Analyze the results and finalize the paper.

## References

[1] S. Ganesh et al., “FSPC: A Lossy Spike Compression Through Correlated-AER Merging in Spiking Neural Networks,” IEEE MCSoC, 2025.